Publication figures and captions

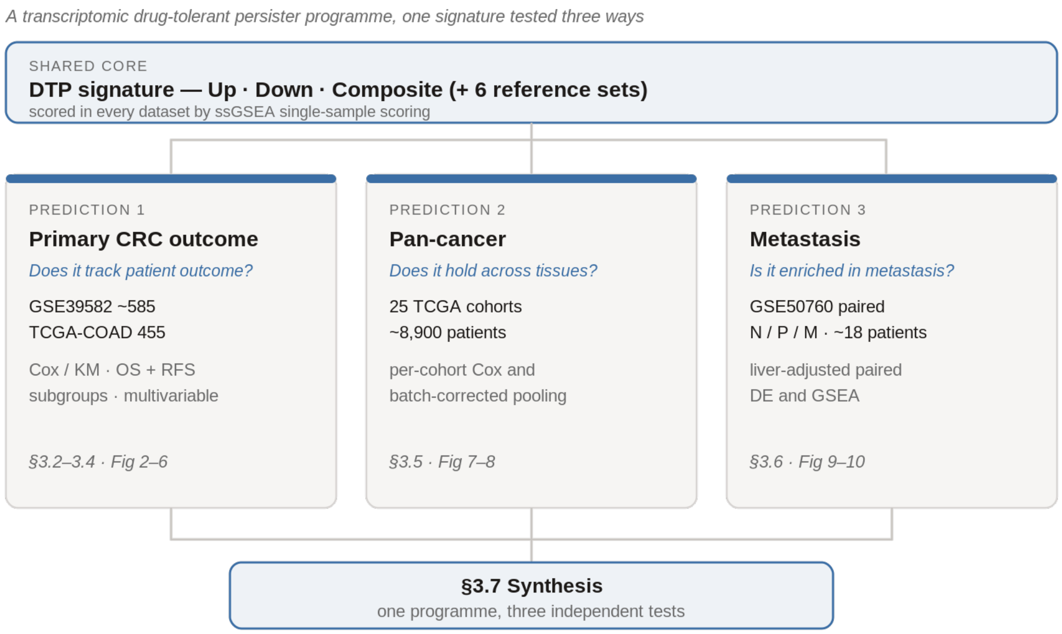

Figure 1

Figure 1. Study design and analysis overview. The drug-tolerant persister signature (Up, Down and Composite scores, with six reference signatures) was scored in every dataset by ssGSEA single-sample scoring, then tested against three predictions of the hypothesis. Prediction 1, association with outcome in primary colorectal cancer, used GSE39582 (about 585 patients) and TCGA-COAD (455), by Cox and Kaplan-Meier analysis of overall survival and recurrence, subgroups and multivariable models (Sections 3.2 to 3.4, Figures 2 to 7). Prediction 2, generalisation across cancer types, used 25 solid-tumour TCGA cohorts (about 8,900 patients), analysed per cohort and as a batch-corrected pool (Section 3.5, Figures 8 and 9). Prediction 3, enrichment in metastasis, used the paired GSE50760 cohort of about 18 patients, by liver-adjusted differential expression and GSEA (Section 3.6, Figures 10 and 11). The three strands are drawn together in Section 3.7.


A
Marisa (GSE39582)
TCGA-COAD
DTP Up
DTP Composite
Foetal intestinal
Regenerative stem cell
Revival revCSC
IBD
DTP Down
MYC module
Columnar stem cell
Whole cohort (3-yr OS)
Whole cohort (3-yr RFS)
Adjuvant-treated (3-yr OS)
Adjuvant-treated (3-yr RFS)
Rank-biserial r
-0.3 -0.2 -0.1 0.0 0.1 0.2 0.3
B
ssGSEA score
Poor
Good
3-year outcome
Deceased / recurred
Alive / recurrence-free

Figure 2

Figure 2. Signature scores versus three-year outcome in colorectal cancer. (A) Effect-size matrix. Each cell shows the Mann-Whitney rank-biserial effect size for a signature score against three-year outcome, for overall survival and recurrence, in the whole cohort and the adjuvant-treated subset of GSE39582 (left) and TCGA-COAD (right). Red marks higher scores in patients who died or recurred within the 36-month landmark, blue the reverse; asterisks mark FDR below 0.05. (B) Score distributions split by three-year outcome, one violin pair per contrast in the same layout, with recurred or deceased patients on the left and recurrence-free or alive patients on the right. Rows are the nine signatures: the DTP up, composite and down scores and the six reference signatures.

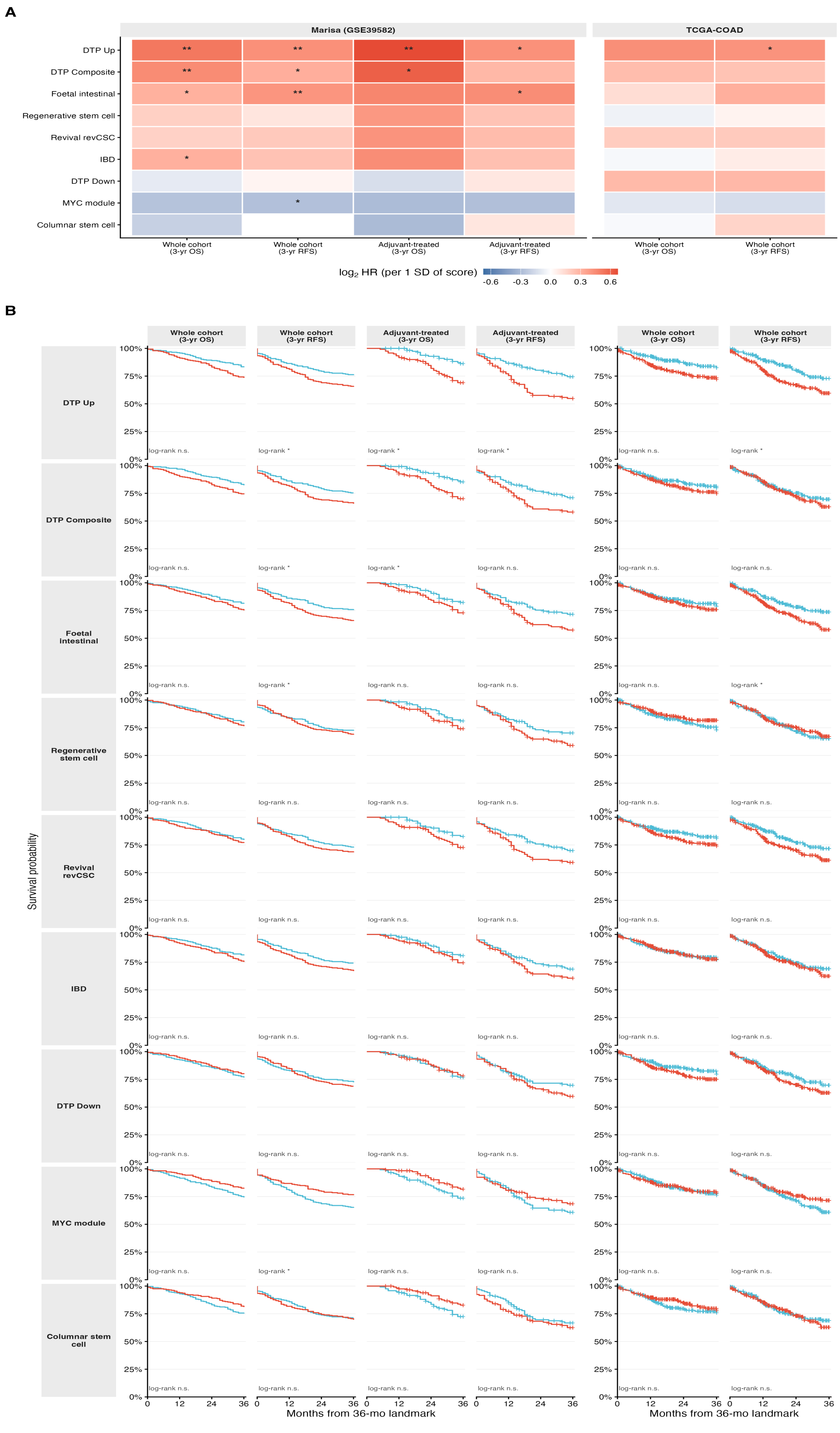
A
Marisa (GSE39582)
TCGA-COAD
DTP Up
DTP Composite
Foetal intestinal
Regenerative stem cell
Revival revCSC
IBD
DTP Down
MYC module
Columnar stem cell
Whole cohort (3-yr OS)
Whole cohort (3-yr RFS)
Adjuvant-treated (3-yr OS)
Adjuvant-treated (3-yr RFS)
log2 HR (per 1 SD of score)
-0.6 -0.3 0.0 0.3 0.6
B
Survival probability
Months from 36-mo landmark
ssGSEA score
Low (< median)
High (> median)
log-rank n.s.
log-rank *

Figure 3

Figure 3. Survival association of signature scores in colorectal cancer. (A) Cox matrix. Each cell shows the hazard ratio per one standard deviation of the continuous score on a log2 scale, from a proportional-hazards model of overall survival or recurrence, in the same layout as Figure 2. Red marks higher hazard with higher score; asterisks mark FDR below 0.05. (B) Kaplan-Meier curves for each signature and contrast, split at the score median, high-score group in red and low-score group in blue. The median split serves display only; inference uses the continuous-score Cox model in panel A. Separation is clearest for the DTP up and composite scores on recurrence and on overall survival in the adjuvant-treated subset.

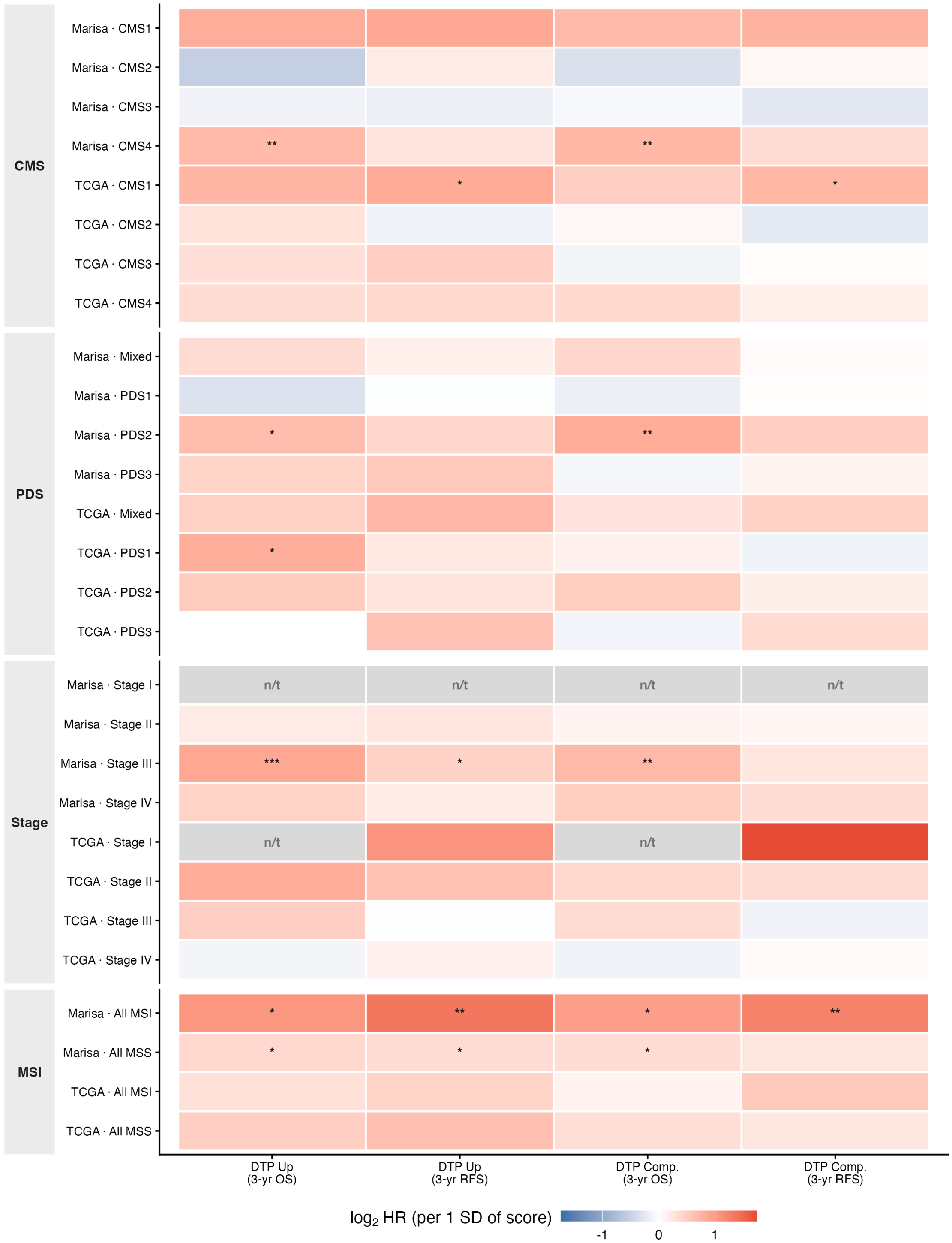

Figure 4

Figure 4. DTP score and outcome within molecular and clinical subgroups. Each cell shows the univariable Cox hazard ratio per one standard deviation of the continuous score on a log2 scale, for the DTP up and composite scores against three-year overall survival and recurrence, within each CMS, PDS, stage and microsatellite stratum of GSE39582 and

TCGA-COAD. Red marks higher hazard with higher score; asterisks mark FDR below 0.05. Grey cells labelled n/t failed the testability gate (fewer than 5 events or 10 patients) and are kept distinct from tested-but-null cells. Most strata are underpowered, so a non-significant or not-testable cell is not evidence of absence. The GSE39582 rows are labelled Marisa, after the source publication.

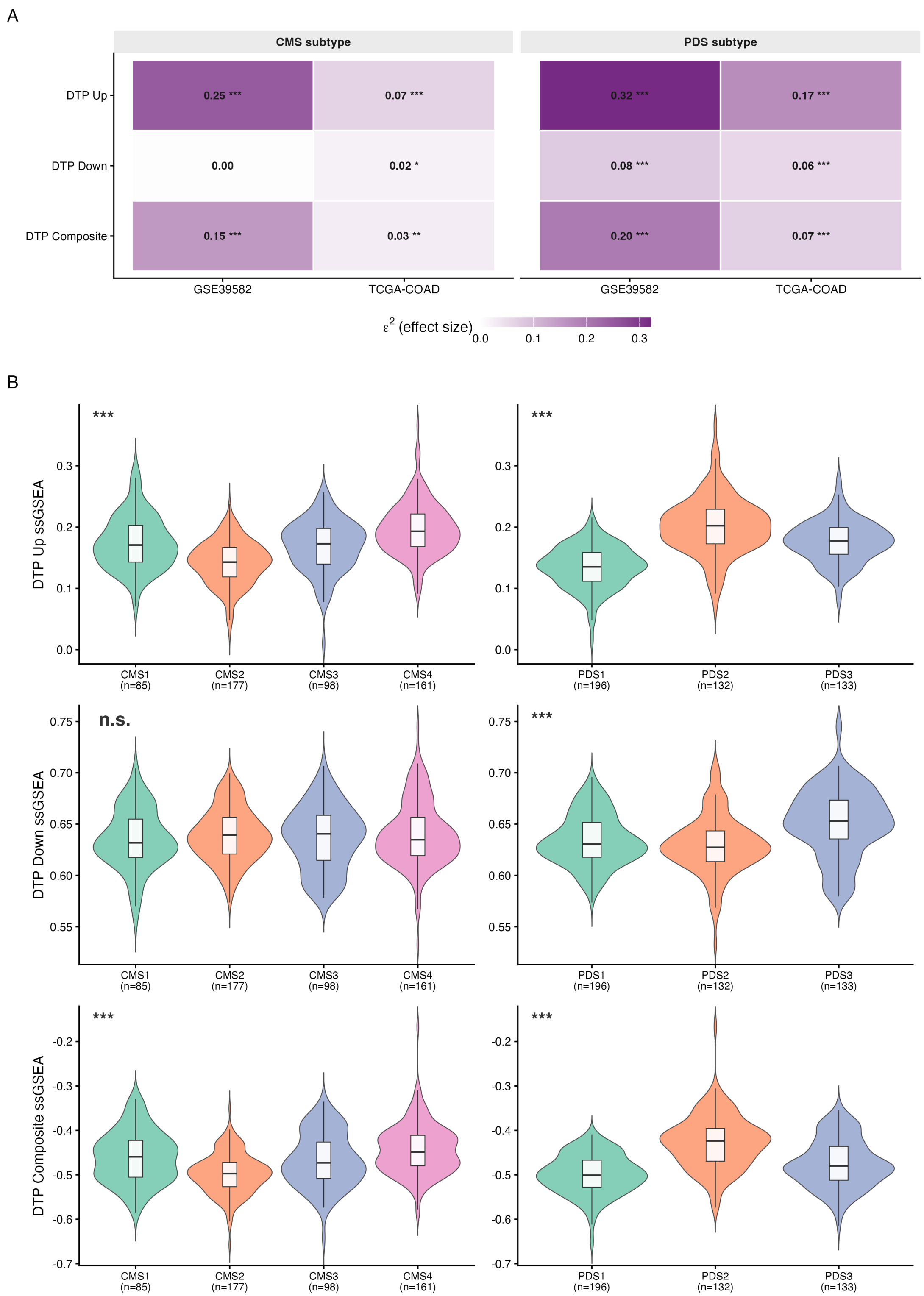
A
CMS subtype
PDS subtype
DTP Up
DTP Down
DTP Composite
0.25 ***
0.07 ***
0.32 ***
0.17 ***
0.00
0.02 *
0.08 ***
0.06 ***
0.15 ***
0.03 **
0.20 ***
0.07 ***
GSE39582
TCGA-COAD
GSE39582
TCGA-COAD
ε² (effect size)
0.0
0.1
0.2
0.3
B
DTP Up ssGSEA
DTP Down ssGSEA
DTP Composite ssGSEA
***
n.s.
CMS1 (n=85)
CMS2 (n=177)
CMS3 (n=98)
CMS4 (n=161)
PDS1 (n=196)
PDS2 (n=132)
PDS3 (n=133)

Figure 5

Figure 5. The DTP score across molecular subtypes. (A) Score-across-subtype effect sizes. Each cell shows the Kruskal-Wallis epsilon-squared effect size for the difference in a score (DTP up, down, composite) across CMS or PDS levels, in GSE39582 and TCGA-COAD; asterisks mark FDR below 0.05. (B) Score distributions across subtype. DTP up, down and composite ssGSEA scores across the CMS1 to CMS4 and PDS1 to PDS3 strata of GSE39582, one violin per subtype level with its sample size.

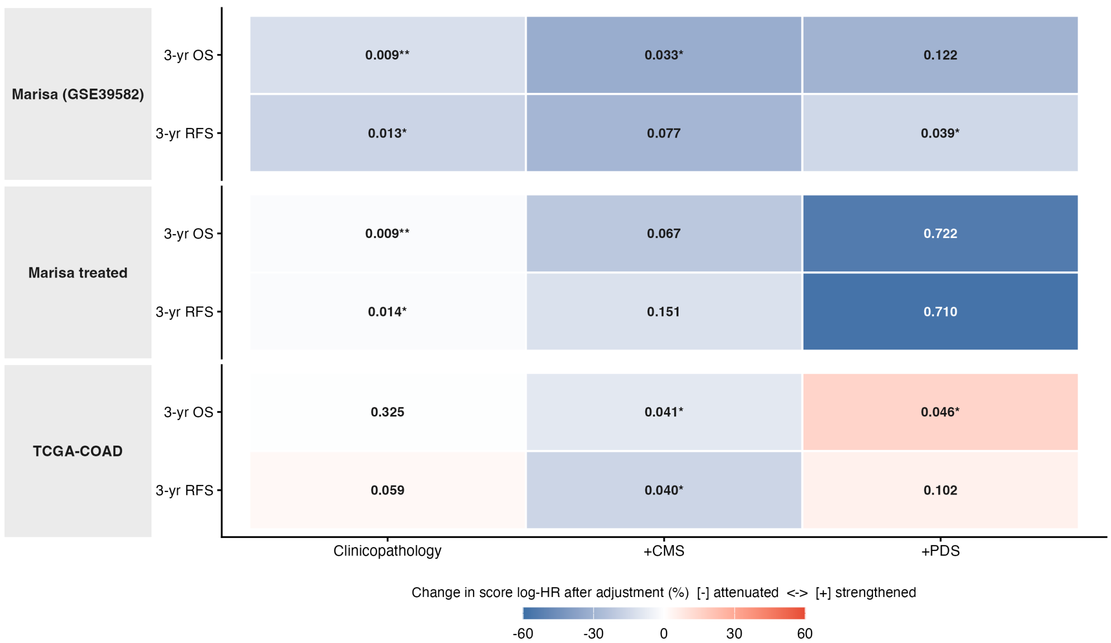

Figure 6

Figure 6. Change in the DTP score effect after adjustment (exploratory). Each cell is the DTP up score term from a landmark Cox model of overall survival or recurrence, adjusted for clinicopathology (stage and microsatellite status), CMS subtype or PDS subtype, in GSE39582 (whole cohort and adjuvant-treated subset) and TCGA-COAD. Tile colour is the percentage change in the score's log hazard ratio after adjustment, relative to the unadjusted model, so blue marks a shift toward the null where the adjustor absorbs part of the signal and red a strengthening; by Greenland's rule of thumb a shift beyond 10% counts as material confounding, so cells within 10% read near-white. The number in each cell is the Benjamini-Hochberg FDR of that score term's likelihood-ratio test, the residual significance after adjustment, and an asterisk marks FDR below 0.05. This panel is exploratory, and adjustment removes only measured confounders.

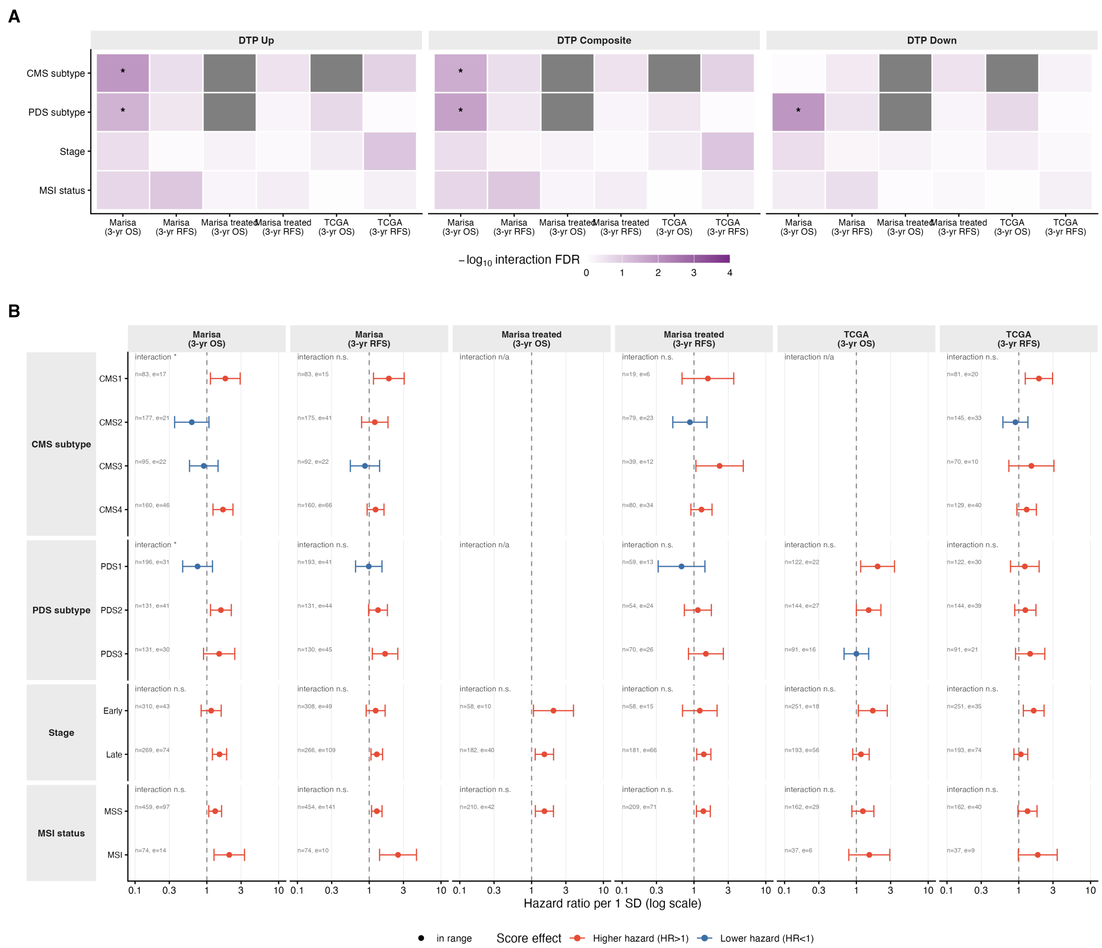

Figure 7

Figure 7. Effect modification and per-subgroup hazard ratios of the DTP score. (A) Interaction map. Each cell shows the -log10 FDR of a likelihood-ratio test comparing a score-by-modifier Cox model against the additive model, for the DTP up, composite and down scores against overall survival and recurrence in GSE39582 and TCGA-COAD, across CMS, PDS, stage and microsatellite modifiers; darker cells and asterisks mark FDR below 0.05. (B) Per-subgroup forest. Continuous-score Cox hazard ratio per one standard deviation of the DTP up score within each subgroup, with 95% confidence interval on a log scale, for both cohorts and both endpoints; red marks a hazard ratio above one and blue below one, and each facet gives its interaction FDR. Effect modification reaches significance only for overall survival in GSE39582, by CMS and PDS.

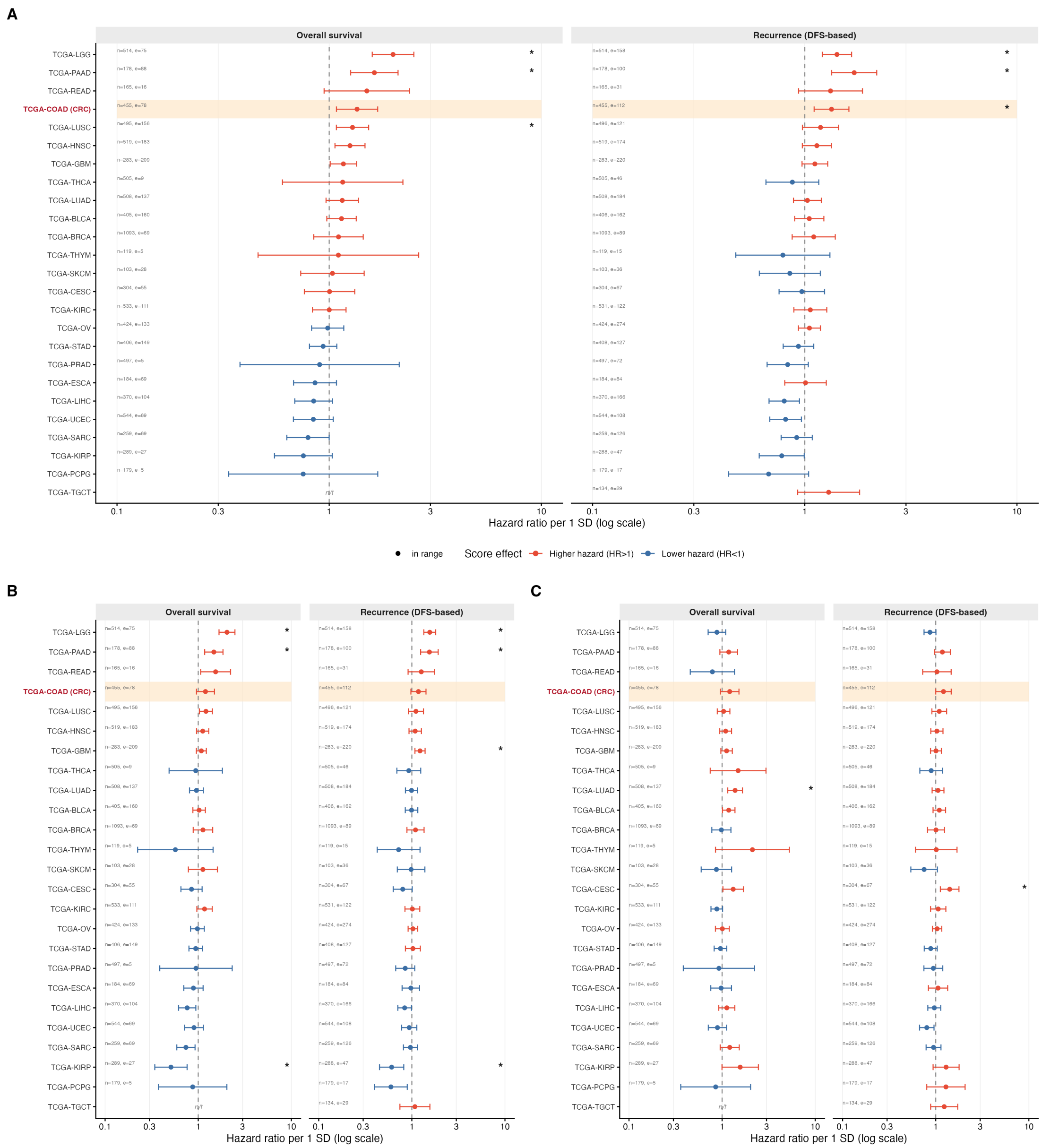

Figure 8

Figure 8. Per-cohort DTP-score survival association across the pan-cancer cohort. Three stacked forests on a shared cohort order, one per score: (A) DTP Up, (B) DTP Composite, (C) DTP Down. Each is the univariable continuous-score Cox hazard ratio per one standard deviation of the score within each of 25 solid-tumour TCGA cohorts, with 95 percent confidence interval on a log axis, faceted by overall survival and recurrence (DFS-based). Red marks a hazard ratio above one and blue below one, an asterisk marks FDR below 0.05, and a grey n/t marks a cohort that failed the events gate (only TCGA-TGCT overall survival). TCGA-COAD, the colorectal tissue, is highlighted. The batch-corrected pooled hazard-ratio forest is provided as an appendix figure.

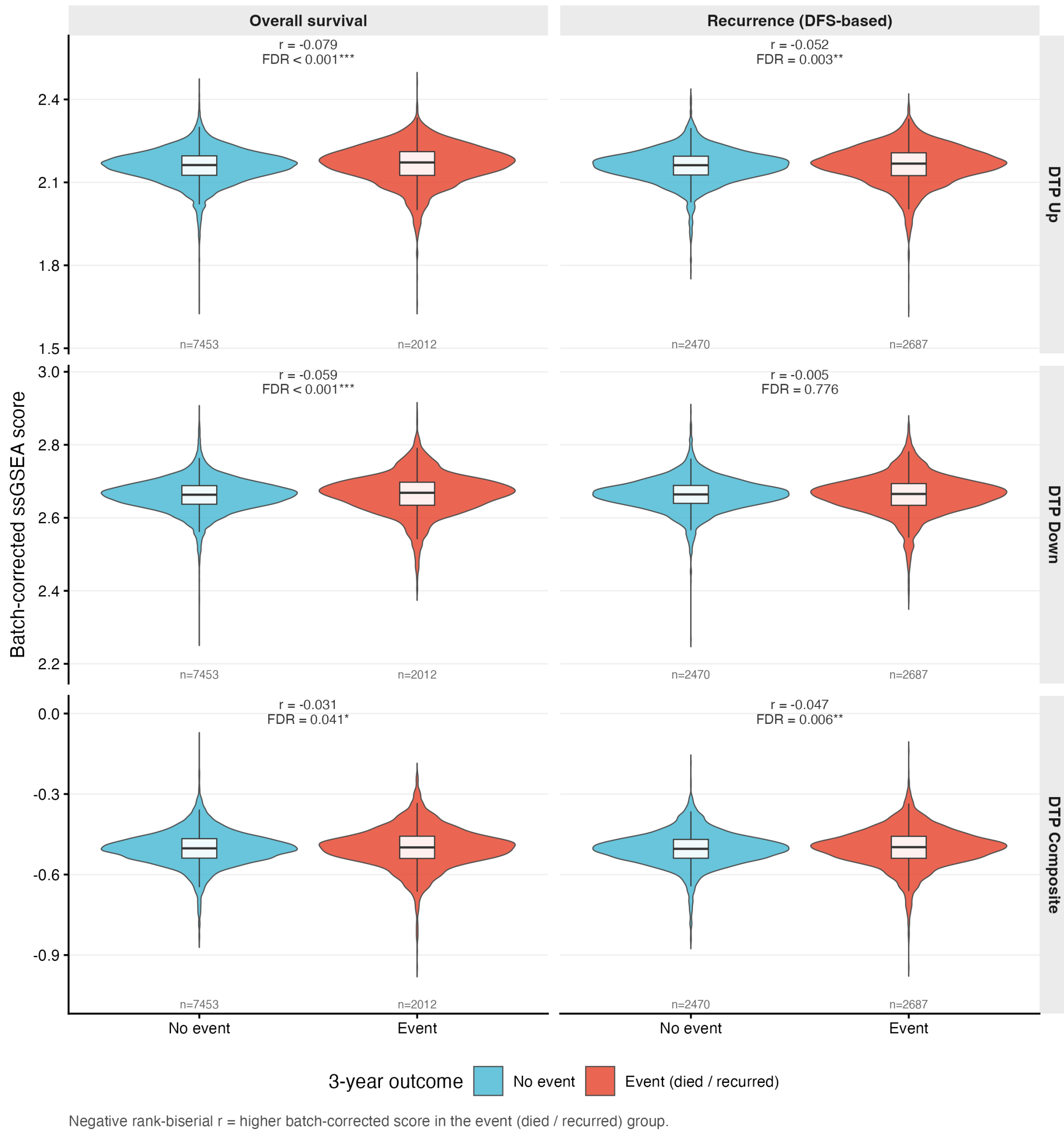

Figure 9

Figure 9. Pooled pan-cancer DTP score by three-year outcome. Batch-corrected score (cancer type as batch) for all 25 cohorts pooled, split by three-year outcome (no event versus died or recurred), for the three scores by the two endpoints. Each panel is annotated with the Wilcoxon rank-biserial effect size and its FDR; a negative sign denotes a higher score in the event group. The distributions overlap almost completely, consistent with the small effect sizes.

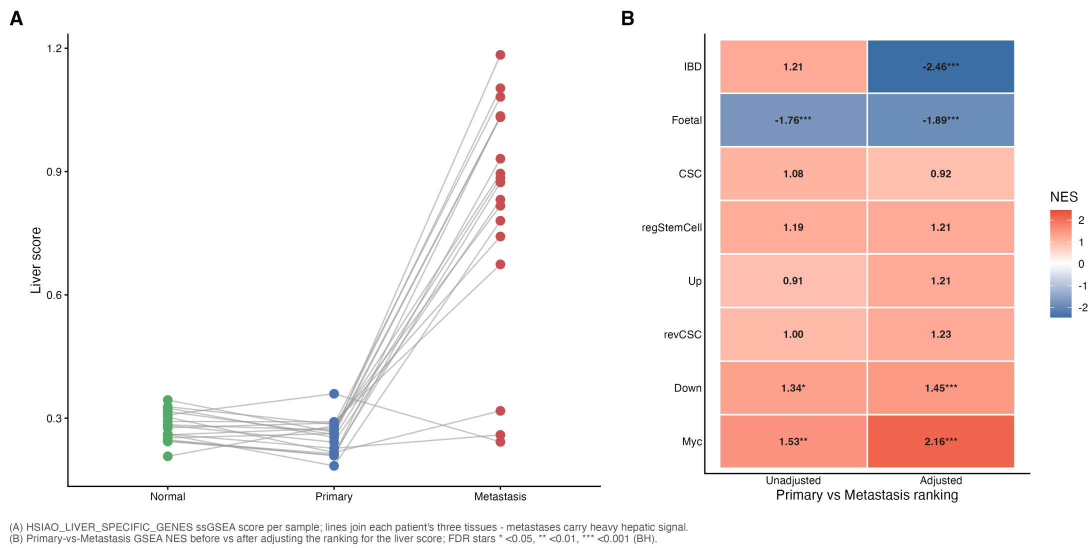

Figure 10

Figure 10. Liver contamination in GSE50760 metastases and its effect on the primary-versus-metastasis enrichment. (A) Per-patient liver score across matched normal, primary and metastasis samples, one line per patient. The liver score is a liver-specific gene score used as a contamination proxy; it is low and flat in normal and primary samples and rises steeply in most metastases, confirming heavy hepatic signal in the metastasis arm. (B) Primary-versus-metastasis GSEA normalised enrichment score for each signature before and after adjusting the ranking for the liver score, with FDR stars. Adjustment flips the inflammatory bowel disease programme from an apparent metastasis enrichment to a primary one, marking it as contamination-driven, and strengthens the Myc, DTP down and foetal signals.

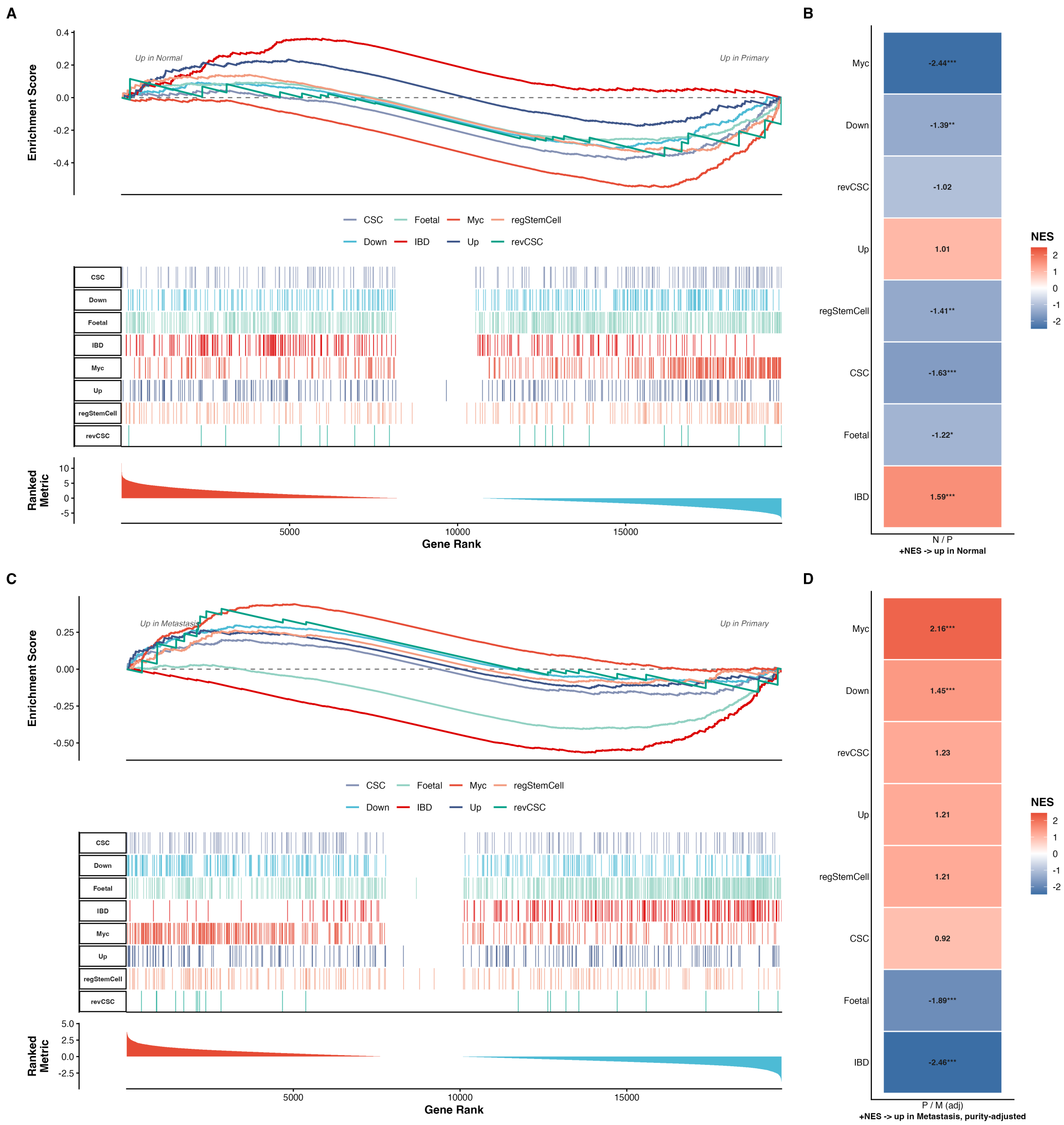

Figure 11

Figure 11. Signature enrichment along the normal to primary to metastasis axis. (A) Normal versus primary running-enrichment curves for all signatures, with the gene-rank ticks and the ranked metric. (B) Normal versus primary NES and FDR heatmap, positive NES meaning up in normal; this contrast is unadjusted because neither arm contains liver. (C) Primary versus metastasis running-enrichment curves. (D) Primary versus metastasis NES and FDR heatmap, positive NES meaning up in metastasis, computed on the liver-adjusted ranking.

METHODS

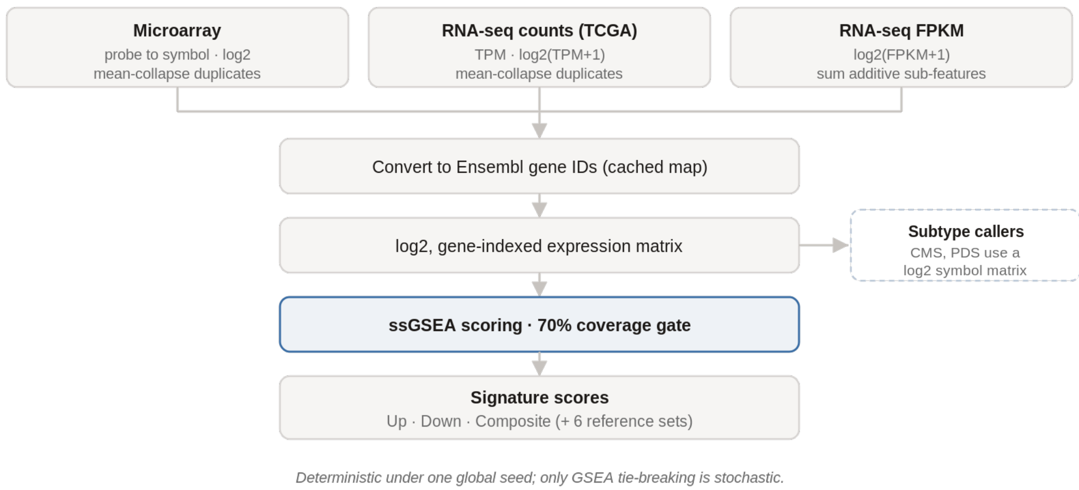

Figure M1

Figure M1. Expression processing and scoring. Each platform was normalised on its own terms, microarray probes collapsed to gene symbols, TCGA counts taken to log2 transcripts per million, and FPKM values to log2 with additive sub-features summed, before all matrices were converted to a single Ensembl-indexed log2 matrix and scored by ssGSEA under a 70 percent gene-coverage gate to yield the three DTP scores and the six reference scores. The CMS and PDS classifiers instead take a log2 gene-symbol matrix (Section 2.7). Scoring is deterministic under the global seed.

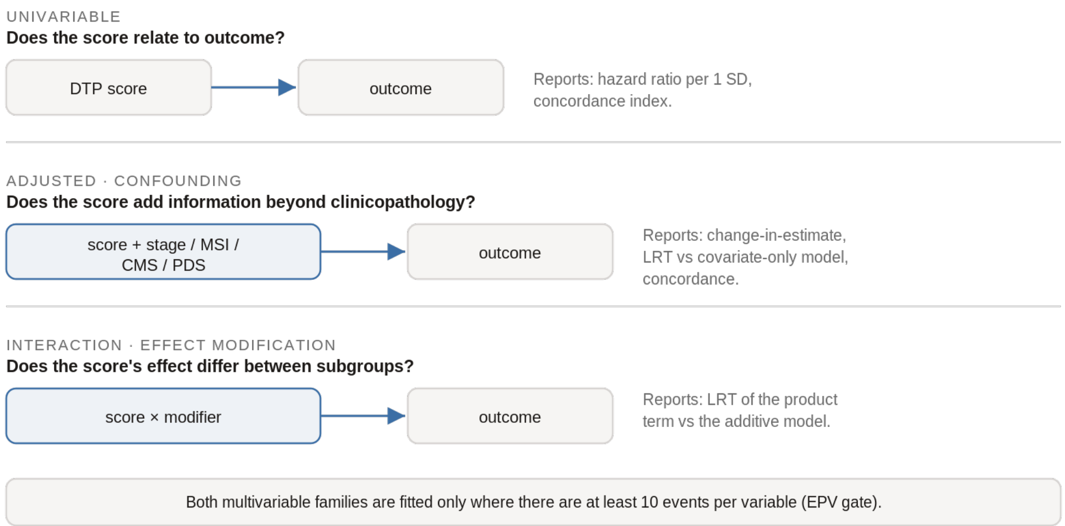

Figure M2

Figure M2. The two Cox families. The univariable model relates the continuous score to outcome. The adjusted (confounding) model adds the score to a covariate-only model and reports the change-in-estimate and the likelihood-ratio test of added fit. The interaction (effect-modification) model tests a score-by-modifier product term against the additive model. Both multivariable families are gated at ten events per variable.